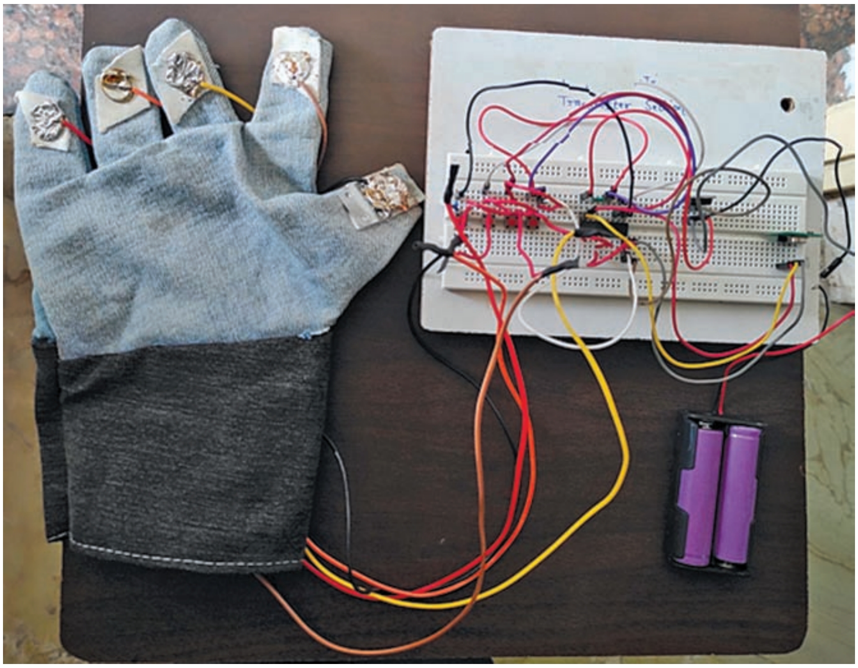

Fig. 5: Transmitter prototype

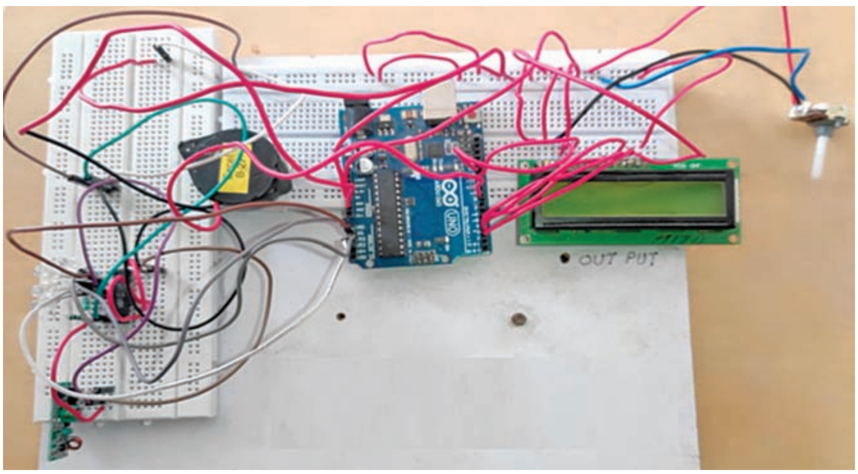

Fig. 6: Receiver prototype

EFY Note
The source code of this project is available for free download at *www.electronicsforu.com*

Upon receiving the signal, the buzzer activates for 5 seconds to notify the user. After this, the buzzer stops, and the requested item's name is displayed on the LCD for 10 seconds.

For demonstration, we programmed four messages: 'BREAKFAST,' 'LUNCH,' 'DINNER,' and 'TEA/COFFEE.' Depending on the switch pressed or the finger touched, one of these messages will appear on the LCD after the 5-second buzzer alert.

Software

The Arduino IDE (integrated development environment) is used for writing, compiling, and uploading the code to the Arduino board. The IDE comes with built-in libraries, including the LiquidCrystal library, which must be installed via the library manager. The analogue pins are then defined as input, and the pins for the LCD are configured accordingly. Fig. 4 shows a code snippet used for the LCD.

EFY note. You can also use an I2C adapter for the display, which requires I2C pins A4 and A5. Using an I2C display simplifies the connection process. If using an I2C display, adjust the code to match the setup.

Construction and testing

Assemble the transmitter circuit as outlined in Fig. 2. See the author's prototype in Fig. 5 for guidance before starting assembly.

In the receiver section, remember to upload the source code to the Arduino Uno before assembly. Once the code is uploaded, assemble the receiver circuit as shown in Fig. 3. Fig. 6 shows the author's prototype of the receiver.

After assembling both the transmitter and receiver, and verifying connections, power on both devices. Test the system by touching the appropriate points on the glove and observing the LCD output while listening for the buzzer sound. EFY

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